Lecture 4: Bifurcations and Qualitative Changes in Dynamics

Modelling Complex Systems (Spring 2026)

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April 7, 2026

Prelecture discussion

Before Lecture 4 I suggest you to take `logisticDiagram.py` and modified it to work for other one parametric family of maps. For example, take

$$f(x) = rx(1-x)(1+\varepsilon(2x-1)^2),$$

for small values of ε (e.g. 0, 10^{-3}, 0.1).

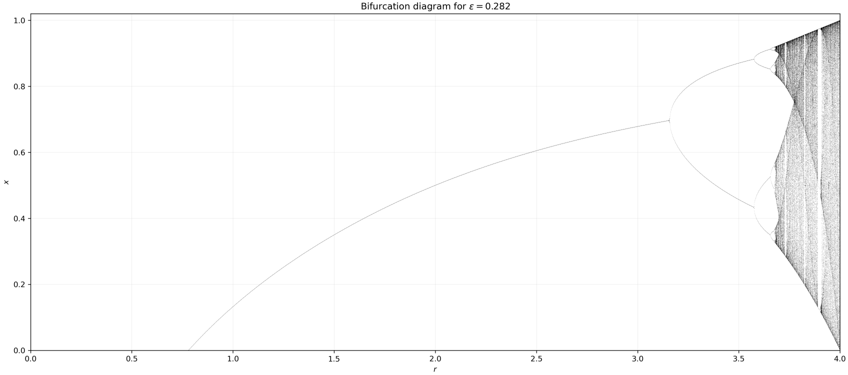
Modified logistic family: $\varepsilon = 0.282$

Bifurcation diagrams for $f(x) = rx(1-x)(1+\varepsilon(2x-1)^2)$

4096 parameter values
1024 transient iterates
512 recorded iterates

Ready for $\varepsilon = 0.282$

Save PNG Save PDF



ε 0.282

Modified logistic family: $\varepsilon = 0.602$

Bifurcation diagrams for $f(x) = rx(1-x)(1+\varepsilon(2x-1)^2)$

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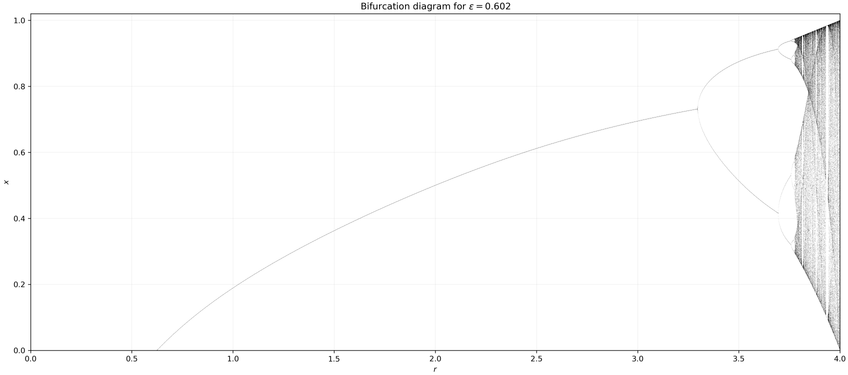

ε 0.602

Modified logistic family: $\varepsilon = 1.000$

Bifurcation diagrams for $f(x) = rx(1-x)(1+\varepsilon(2x-1)^2)$

4096 parameter values
1024 transient iterates
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Ready for $\varepsilon = 1.000$

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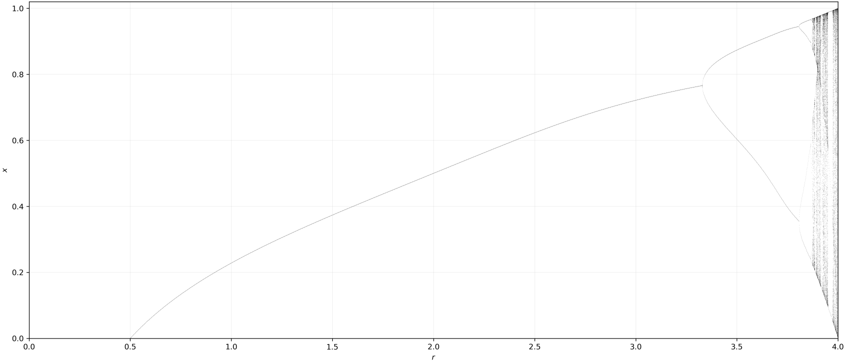
Bifurcation diagram for $\varepsilon = 1$



ε 1

Lecture goals

- Define what is meant by a parameter-dependent dynamical system.
- Explain what a bifurcation is.
- Relate bifurcations to stability changes of fixed points and periodic orbits.
- Study the main local bifurcations for one-dimensional discrete maps.
- Connect the theory with the logistic map and with numerical exploration of the standard map.

Why parameters matter

A parameter-dependent system

In applications, a dynamical system is rarely given by a single fixed rule.

We usually study a family of systems depending on a parameter μ:

$$x_{n+1} = f_\mu(x_n).$$

The parameter may represent:

- a reproduction rate,
- a coupling strength,
- an external forcing,
- a discretization or control parameter.

Qualitative versus quantitative change

Not every change in a parameter is equally important.

- A **quantitative change** modifies numerical values while preserving the same type of long-time behavior.
- A **qualitative change** modifies the structure of the dynamics itself.

Examples of qualitative change:

- a stable equilibrium becomes unstable,
- a new equilibrium appears,
- a stable periodic orbit is created,
- ordered motion gives way to more complicated behavior.

Definition of bifurcation

Informal definition

A **bifurcation** occurs when a small change in a parameter produces a qualitative change in the long-time dynamics.

What may change at a bifurcation value $\mu = \mu_c$?

- the number of fixed points,
- the stability of a fixed point,
- the existence of periodic orbits,
- the geometry of invariant sets.

What do we track as the parameter varies?

Typical objects of interest are:

- fixed points $x^*(\mu)$,
- periodic orbits,
- their stability,
- the observed long-time attractors.

In practice, we often combine:

- analytic calculations,
- local linearization,
- numerical continuation,
- direct simulation.

Local viewpoint

What should local dynamics near a fixed point look like?

Suppose x^* is a fixed point of

$$x_{n+1} = f_\mu(x_n).$$

If we start with x_0 close to x^*, we would like to understand:

- whether the orbit moves toward or away from x^*,
- whether it does so monotonically or with oscillation,
- which simple model should describe the motion nearby.

This is the basic local question behind bifurcation theory.

A theorem to keep in mind: Hartman-Grobman theorem

For a **hyperbolic** fixed point ($|f_\mu'(x)| \neq 1$), the local dynamics is as simple as the linearization.

Informally, if x^* is a fixed point and

$$|f_\mu'(x^*)| \neq 1,$$

then near x^* the nonlinear map is locally conjugate to its linear part.

So, after a suitable local change of coordinates, the dynamics looks like

$$y_{n+1} = \lambda y_n, \qquad \lambda = f_\mu'(x^*).$$

Why is this important?

This theorem explains why the derivative already tells us the local picture.

For the linear map

$$y_{n+1} = \lambda y_n,$$

we know immediately:

- if $|\lambda| < 1$, iterates contract toward 0,
- if $|\lambda| > 1$, iterates move away from 0,
- if $\lambda < 0$, the orbit alternates sides while doing so.

Hence hyperbolic fixed points have a locally robust qualitative behavior.

Question before the theorem

Questions for discussion

If a fixed point is hyperbolic for one parameter value μ_0, what should happen when we change the parameter slightly?

Should we expect:

- the fixed point to disappear immediately,
- a nearby fixed point to remain,
- its stability type to remain the same,
- or some combination of these?

Discuss this first from the fixed-point equation

$$f_\mu(x) = x.$$

Why hyperbolic fixed points persist

Write the fixed-point problem as

$$F(\mu, x) = f_\mu(x) - x = 0.$$

At a hyperbolic fixed point (μ_0, x^*) we have

$$\partial_x F(\mu_0, x^*) = f'_{\mu_0}(x^*) - 1 \neq 0.$$

Then the **Implicit Function Theorem** gives a nearby branch

$$x^*(\mu)$$

of fixed points for μ close to μ_0.

So hyperbolic fixed points do not suddenly vanish under small parameter perturbations.

Why this matters for bifurcations

If hyperbolic fixed points persist and keep their local type, then a bifurcation should not occur while hyperbolicity is preserved.

Therefore, to look for a qualitative change, we are naturally led to the *non-hyperbolic* threshold

$$|f_\mu'(x^*)| = 1.$$

This is why the condition $|f_\mu'(x^*)| = 1$ plays such a central role in one-dimensional bifurcation theory.

Beyond the Implicit Function Theorem

What if $f'_\mu(x^*) = -1$?

Questions for discussion

Consider a fixed point with

$$f'_\mu(x^*) = -1.$$

- Is this fixed point hyperbolic?
- Does the local conjugacy theorem for hyperbolic fixed points still apply?
- Can we still use the Implicit Function Theorem on

$$F(\mu, x) = f_\mu(x) - x?$$

- If the fixed point persists, what kind of qualitative change might still occur?

When the Implicit Function Theorem fails

To understand what may happen, it is useful to forget dynamics for a moment and think geometrically.

Consider an equation

$$F(x, y) = 0$$

near the point $(0, 0)$.

If

$$\partial_y F(0, 0) \neq 0,$$

then the Implicit Function Theorem says that nearby the solution set is the graph of a function

$$y = y(x).$$

But what should we expect if

$$\partial_y F(0, 0) = 0?$$

Possible local pictures for $F(x, y) = 0$

If the Implicit Function Theorem fails for solving for y, the set $F(x, y) = 0$ may look very different near $(0, 0)$.

Possible local pictures include:

- a vertical tangent, so the set is not the graph of $y = y(x)$,
- two crossing branches,
- two branches touching tangentially,
- a cusp or another singular shape,
- an isolated point.

So failure of the theorem does not tell us exactly what happens, but it tells us that the simple graph picture can break down.

A simple example

Consider

$$F(x, y) = y^2 - x.$$

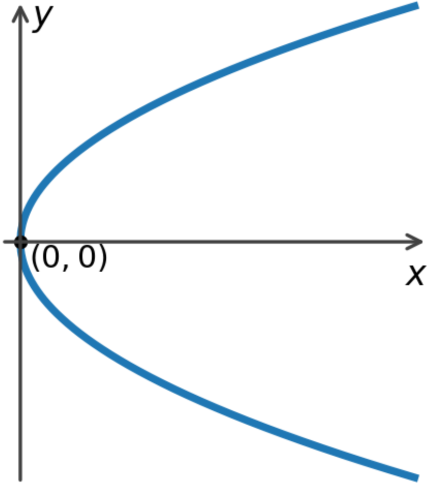
At $(0, 0)$ we have

$$F(0, 0) = 0, \qquad \partial_y F(0, 0) = 0.$$

The solution set is

$$y = \pm\sqrt{x}.$$

Near $(0, 0)$ this is not one branch $y = y(x)$, but two branches meeting at a point.



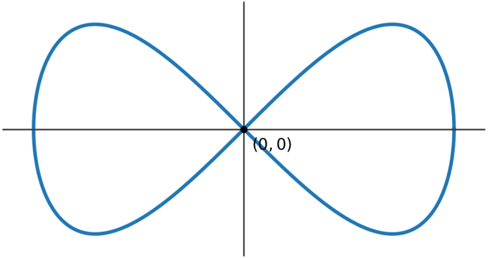
Another example: the lemniscate

Another typical picture is a self-intersection.

For example, the lemniscate of Gerono can be written as

$$F(x, y) = x^4 - x^2 + y^2 = 0.$$

At the crossing point, the solution set is not a single graph $y = y(x)$.



Bonus: How to compute at a singular point

What is a singular point?

Consider an implicit curve

$$F(x,y)=0.$$

A point $(0,0)$ is called a **singular point** if

$$F(0,0)=0, \qquad \partial_x F(0,0)=0, \qquad \partial_y F(0,0)=0.$$

At such a point, the usual Implicit Function Theorem does not help us solve locally for

$$y=y(x) \qquad \text{or} \qquad x=x(y).$$

So we need another way to detect the local branches.

A useful idea: polar coordinates

Around $(0,0)$, write

$$x = \rho\cos\theta, \qquad y = \rho\sin\theta.$$

Then substitute into the equation:

$$F(\rho\cos\theta, \rho\sin\theta) = 0.$$

After that, remove as many powers of ρ as possible.

The remaining equation often reveals the allowed directions θ of the branches through the singular point.

Why this helps

Near the origin, the smallest power of ρ usually contains the first nontrivial geometric information.

After factoring out the largest common power of ρ, we often obtain an equation of the form

$$G(\rho, \theta) = 0,$$

with a meaningful limit as $\rho \to 0$.

Setting $\rho = 0$ in that reduced equation gives candidate angles

$$\theta = \theta_0$$

for the tangent directions of the branches.

If the reduced equation is still degenerate, we repeat the same idea again.

Applying the method to the Gerono lemniscate

Recall

$$F(x, y) = x^4 - x^2 + y^2.$$

At $(0, 0)$,

$$F(0, 0) = 0, \qquad \partial_x F(0, 0) = 0, \qquad \partial_y F(0, 0) = 0,$$

so this is a singular point.

Now write

$$x = \rho \cos \theta, \qquad y = \rho \sin \theta.$$

Desingularizing the Gerono equation

Substituting polar coordinates gives

$$r^4 \cos^4 \theta - r^2 \cos^2 \theta + r^2 \sin^2 \theta = 0.$$

Factor out the largest common power of r:

$$r^2 \left(r^2 \cos^4 \theta - \cos^2 \theta + \sin^2 \theta \right) = 0.$$

So for $r \neq 0$ we reduce to

$$r^2 \cos^4 \theta - \cos^2 \theta + \sin^2 \theta = 0.$$

Now let $\rho \to 0$.

Angles and branches for the Gerono lemniscate

At leading order we obtain

$$-\cos^2\theta + \sin^2\theta = 0.$$

Equivalently,

$$\sin^2\theta = \cos^2\theta, \qquad \tan^2\theta = 1.$$

Hence the possible directions are

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}.$$

These are the two crossing lines

$$y = x \qquad \text{and} \qquad y = -x.$$

How many branches do we have?

In this example, the reduced equation gives four solutions for θ in $[0, 2\pi)$.

These correspond to the local directions of the curve at the singular point.

So the number of θ-solutions tells us the number of local branches passing through $(0, 0)$.

For the Gerono lemniscate:

- there are four angle solutions,
- arranged into two opposite pairs,
- therefore we see two smooth branches crossing at the origin.

Study of the period doubling

Why use f^2 for period doubling?

For the logistic map

$$f_r(x) = rx(1-x),$$

a period-2 orbit of f_r is a fixed point of the second iterate:

$$f_r^2(x) = x.$$

So near the period-doubling point $r = 3$, it is natural to study

$$F(r,x) = f_r^2(x) - x.$$

The relevant fixed point is

$$x^* = 1 - \frac{1}{r}, \qquad x^*(3) = \frac{2}{3}.$$

A singular point for $f_r^2(x) - x$

Introduce local coordinates around the bifurcation point:

$$a = r - 3, \qquad y = x - \frac{2}{3}.$$

Then we study

$$G(a, y) = f_{3+a}^2\left(\frac{2}{3} + y\right) - \left(\frac{2}{3} + y\right).$$

At the origin,

$$G(0,0) = 0, \qquad \partial_a G(0,0) = 0, \qquad \partial_y G(0,0) = 0.$$

So $(a, y) = (0, 0)$ is a singular point of the curve

$$G(a, y) = 0.$$

Why second order is enough

Since $(a, y) = (0, 0)$ is a singular point, the constant and linear terms vanish:

$$G(0,0) = 0, \qquad \partial_a G(0,0) = 0, \qquad \partial_y G(0,0) = 0.$$

Therefore the Taylor expansion starts at order 2:

$$G(a, y) = \frac{1}{2} G_{aa}(0,0)a^2 + G_{ay}(0,0)ay + \frac{1}{2} G_{yy}(0,0)y^2 + O(3).$$

So to detect the local branches, we only need the quadratic part of $f_r^2(x) - x$.

How to get the quadratic coefficients

Let

$$H(r,x) = f_r^2(x).$$

Then

$$G(a,y) = H\left(3+a, \frac{2}{3}+y\right) - \left(\frac{2}{3}+y\right).$$

So the quadratic Taylor coefficients of G come from derivatives of H at

$$(r,x) = \left(3, \frac{2}{3}\right).$$

The quadratic coefficients

The needed second derivatives are

$$G_{aa}(0,0) = -\frac{4}{9}, \qquad G_{ay}(0,0) = 2, \qquad G_{yy}(0,0) = 0.$$

Therefore the quadratic part is

$$\frac{1}{2}G_{aa}(0,0)a^2 + G_{ay}(0,0)ay + \frac{1}{2}G_{yy}(0,0)y^2 = -\frac{2}{9}a^2 + 2ay.$$

This is exactly the order-2 information used in the polar-coordinate analysis.

Why do the other terms vanish?

First,

$$G_y(0,0) = \partial_x H(3, 2/3) - 1 = (f_3'(2/3))^2 - 1 = (-1)^2 - 1 = 0,$$

so there is no linear term in y.

Also,

$$G_{yy}(0,0) = \partial_{xx} H(3, 2/3) = (f_3^2)''(2/3) = 0,$$

because at a fixed point of multiplier -1,

$$(f^2)''(x^*) = f''(x^*)((f'(x^*))^2 + f'(x^*)) = 0.$$

Polar coordinates near the period doubling

Write

$$a = \rho \cos\theta, \qquad y = \rho \sin\theta.$$

Expanding $G(a, y)$ near $(0, 0)$ gives

$$G(a, y) = -\frac{2}{9}a^2 + 2ay + O\Big((|a| + |y|)^3\Big).$$

Hence

$$G(\rho\cos\theta, \rho\sin\theta) = \rho^2 \left(-\frac{2}{9}\cos^2\theta + 2\cos\theta\sin\theta\right) + O(\rho^3).$$

Removing the common factor ρ^2, the leading angular equation is

$$-\frac{2}{9}\cos^2\theta + 2\cos\theta\sin\theta = 0.$$

Directions of the branches

The reduced equation becomes

$$\cos\theta\left(\sin\theta - \frac{1}{9}\cos\theta\right) = 0.$$

So the possible directions are:

- $\cos\theta = 0$, that is
$$\theta = \frac{\pi}{2}, \frac{3\pi}{2},$$
- or
$$\tan\theta = \frac{1}{9},$$
which gives the opposite pair of directions on the line
$$y = \frac{1}{9}a.$$

Thus the singular point has two different types of local branches.

Interpretation for the logistic map

The line

$$y = \frac{1}{9}a$$

is the tangent direction of the fixed-point branch

$$x^*(r) = 1 - \frac{1}{r}.$$

The vertical direction

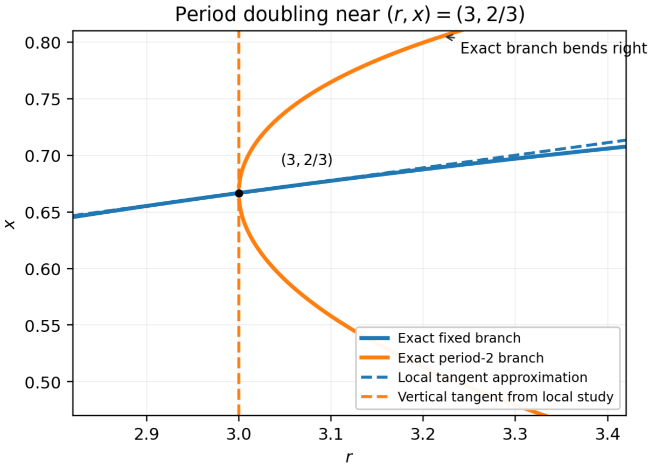
$$a = 0$$

corresponds to the new period-2 branch born at the bifurcation.

So the polar-coordinate computation detects exactly what happens at period doubling:

- one branch is the old fixed point,
- the other local directions belong to the emerging period-2 orbit.

Local picture near the period doubling



Computing the derivatives with Python

A minimal sympy script can compute the derivatives for us:

```
import sympy as sp

r, x, a, y = sp.symbols('r x a y')
f = r * x * (1 - x)
f2 = sp.expand(f.subs(x, f))
G = sp.expand(f2.subs({r: 3 + a,
                       x: sp.Rational(2, 3) + y})
              - (sp.Rational(2, 3) + y))

print(sp.diff(G, a, 2).subs({a: 0, y: 0}))
print(sp.diff(G, a, y).subs({a: 0, y: 0}))
print(sp.diff(G, y, 2).subs({a: 0, y: 0}))
```

SymPy output

Stored in:

`scriptLogisticSymbolicManipulation/logisticPeriodDoublingSympy.py`

Output:

$$G_{aa}(0,0) = -\frac{4}{9}, \qquad G_{ay}(0,0) = 2, \qquad G_{yy}(0,0) = 0.$$

Next meeting: computer lab

Next time we meet

Next time we meet, it will be in a computer lab.

There we will:

- start working on the projects,
- finalize the creation of groups,
- discuss practical questions about the project work.

Students can also bring their own computer if they want.

Summary

Take-away points

- A bifurcation is a qualitative change in long-time dynamics caused by a parameter variation.
- Hyperbolic fixed points are locally conjugate to their linearization and persist under small parameter perturbations.
- When the Implicit Function Theorem fails, singular points can produce branching, crossing, or creation of new solution curves.
- A useful way to study a singular point of $F(x, y) = 0$ is to pass to polar coordinates (ρ, θ) and factor out the lowest power of ρ.
- For the Gerono lemniscate, the number of angle solutions gives the number of local branches through the singular point.
- For the logistic map, period doubling at $r = 3$ is detected by studying $f_r^2(x) - x$ near $(r, x) = (3, 2/3)$.

Suggested reading

For students who want to read more:

- **Strogatz, *Nonlinear Dynamics and Chaos***: read the chapter on one-dimensional maps, especially fixed points, cobweb plots, bifurcation diagrams, the logistic map, and period doubling.
- **Hirsch, Smale, Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos***: read the chapters/sections on discrete dynamical systems, hyperbolic fixed points, and Hartman–Grobman / local linearization.
- **Devaney, *An Introduction to Chaotic Dynamical Systems***: read the chapters on iteration of interval maps, periodic points, and the logistic family.
- **Kuznetsov, *Elements of Applied Bifurcation Theory***: for stronger students, read the sections on local bifurcations of one-dimensional maps and period-doubling (flip) bifurcation.

Chapter numbering may vary slightly by edition, so use the chapter titles above as the main guide.